Middle East Technical University
Department of Electrical and Electronics Engineering

# EE201 – Circuit Theory I

## Lecture 10: LTI Resistive Circuits and Equivalent One-Ports

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| Colour code used in these notes and in the slides | | |
|---|---|---|
| current $i$ | voltage $v$ | a part of a circuit (box $N$), terminals $a$, $b$ |

**Why this lecture?** So far, KCL and KVL were written for any circuit, with any elements. From now on we restrict the elements. In an **LTI resistive circuit** every equation is linear and algebraic. Such a circuit can be solved in two ways: by transforming it into a simpler circuit, or by writing the equations of the circuit as it is. This lecture gives the tools for the first way: the substitution theorem, one-ports and their equivalents, series and parallel connections, and the exchange of a voltage source and a current source.

## 1. LTI resistive circuits

First classify the elements, then the circuit. Independent sources are left aside: they are the inputs (see below).

### Resistive and dynamic elements

- A **resistive** element relates its voltages and currents at the same instant by an algebraic law; it has no memory. Examples: resistors, diodes, dependent sources, ideal transformers, and the ideal op-amp met later. "Resistive" covers all of them, not only resistors.
- A **dynamic** element has a derivative in its law, so it has memory: capacitors ($i = C\,dv/dt$), inductors, coupled inductors.
- An element is **linear** if its law is linear in its voltages and currents, and **time-invariant** if its law does not change with time.

### Nonlinear, time-varying, dynamic circuits

- A **nonlinear** circuit contains at least one nonlinear element, e.g. a diode. Otherwise it is **linear**.
- A **time-varying** circuit contains at least one time-varying element, e.g. $R(t)$. Otherwise it is **time-invariant**.
- A **dynamic** circuit contains at least one dynamic element: an $L$ or a $C$. Otherwise it is **resistive**.

An **LTI resistive circuit** is linear, time-invariant and resistive: apart from the independent sources, every element is linear, time-invariant and resistive (LTI resistors, linear dependent sources, ideal transformers, ...).

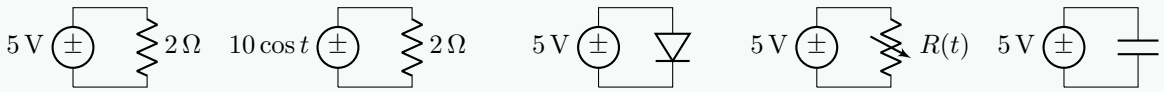
### Independent sources are inputs

An independent source is an **external input**: a given signal of time, such as $v_s(t) = 10\cos t$ V. The three pairs of words describe the circuit that the inputs drive, not the inputs.

**What varies in time?** In every circuit the voltages and currents are functions of time. What matters is the *element law*. A time-invariant resistor obeys $v(t) = R\,i(t)$ with the same $R$ at every instant; a time-varying one obeys $v(t) = R(t)\,i(t)$: the law itself changes. Linear and nonlinear also refer to the element law.

**Example 10.1 — which circuit is which?**

Each circuit is a source and one more element.



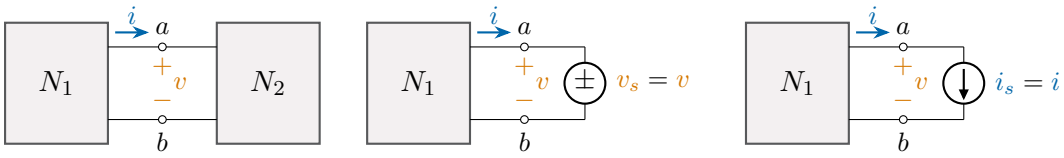

(a) LTI resistive (b) LTI resistive (c) nonlinear (diode) (d) time-varying (e) dynamic

(b) is LTI resistive although its source varies in time. In (d), $R(t) = 2 + \sin t\ \Omega$. (c) and (d) are still resistive; (e) is still linear and time-invariant.

**What we gain.** In an LTI resistive circuit KCL and KVL are linear, and every element equation is linear and algebraic: $v = Ri$ for a resistor, $v = \mu v_x$ or $i = g\,v_x$ for a dependent source, $v_2 = nv_1$ and $i_1 = -ni_2$ for an ideal transformer. No derivative appears. Time enters only through the values of the independent sources, so at each instant the circuit is solved like a circuit with constant sources.

## 2. The substitution theorem

Split a circuit into two parts $N_1$ and $N_2$, joined at two terminals $a$ and $b$. Let $v(t)$ be the voltage between $a$ and $b$, and $i(t)$ the current from $N_1$ into $N_2$ at $a$.




(I) the circuit (II) $N_2 \to$ voltage source (III) $N_2 \to$ current source

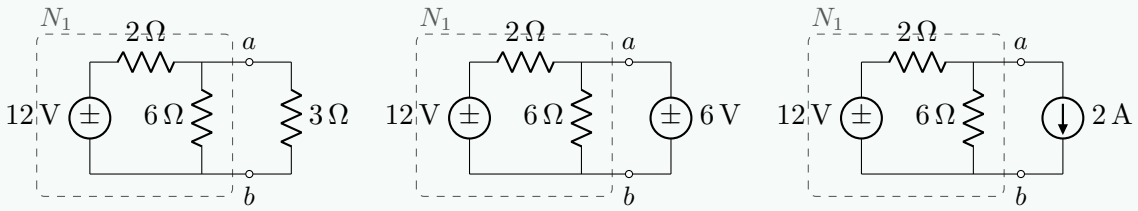
**Substitution theorem**

Replace $N_2$ by an independent voltage source $v_s(t) = v(t)$, or by an independent current source $i_s(t) = i(t)$. No branch voltage or branch current inside $N_1$ changes. An observer inside $N_1$ cannot tell (I), (II) and (III) apart.

**Why it works.** The equations inside $N_1$ (its KCL, KVL and element equations) do not change. $N_1$ meets the rest of the circuit only through $v$ and $i$. The source fixes one of them at the value it had in (I), and the solution of (I) still satisfies every equation of (II) and (III). $N_2$ can be anything, even nonlinear: only its port values matter.

**One condition.** The new circuit must have a unique solution. If $N_1$ is a single current source, replacing $N_2$ by a current source puts two current sources in series: their currents are equal, but the voltage $v$ is no longer fixed.

**Example 10.2 — substitution in a small circuit**




(I) $N_2$: the 3 Ω resistor (II) a 6 V source (III) a 2 A source

**(I)** The 6 Ω and 3 Ω are in parallel: 2 Ω. The two 2 Ω share the 12 V, so $v = 6$ V, and $i = 6/3 = 2$ A flows into $N_2$. Inside $N_1$: the 2 Ω carries $(12-6)/2 = 3$ A and the 6 Ω carries $6/6 = 1$ A.

**(II)** The 6 V source fixes $v = 6$ V. Again the 2 Ω carries 3 A and the 6 Ω carries 1 A; the source takes the remaining 2 A.

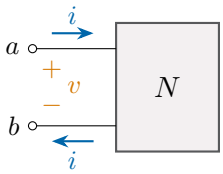
**(III)** The 2 A source takes 2 A from node $a$. KCL at $a$: $(12-v)/2 = v/6 + 2$, so $v = 6$ V, and the currents inside $N_1$ are 3 A and 1 A again.

The theorem lets us cut a circuit and study one part at a time. It is also the key step in later proofs, where a complicated part is replaced by a source.

## 3. Ports and equivalent one-ports

**Port, one-port**

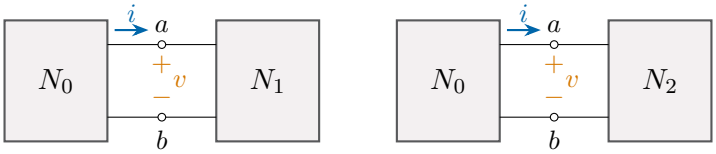
A pair of terminals is a **port** if the current entering one terminal equals the current leaving the other. A **one-port** is a circuit, or a circuit element, with one pair of terminals accessible from outside.




A one-port is always a port: by KCL on a closed surface around $N$, the current $i$ that enters at $a$ leaves at $b$. We take $i$ entering the + terminal of $v$. The relation between $v$ and $i$ is the $i$–$v$ **characteristic** of the one-port. A resistor is the simplest one-port: $v = Ri$.

**Equivalent one-ports**

Two one-ports $N_1$ and $N_2$ are **equivalent**, $N_1 \equiv N_2$, if they have the same $i$–$v$ characteristic at their terminals. Then, for any one-port $N_0$ connected to them, every branch voltage and branch current inside $N_0$ is the same with $N_1$ or with $N_2$.

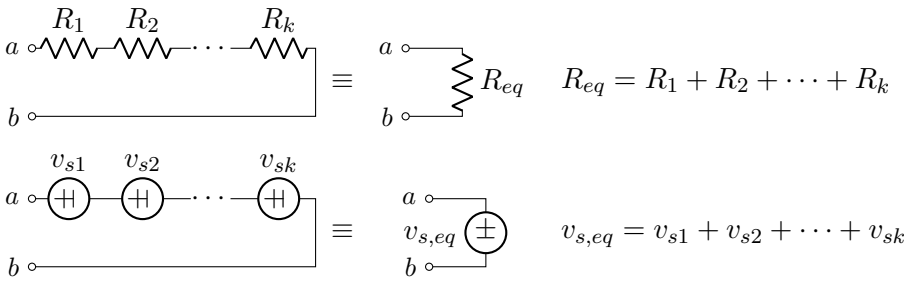



$N_0$ meets the rest only through $v$ and $i$. The equations of the whole circuit are those of $N_0$ plus the $i$–$v$ characteristic at the port. Equivalent one-ports give the same equations, so the same solution inside $N_0$. Inside $N_1$ and $N_2$ the branch values may differ.

## 4. Series and parallel connections

### Series connection

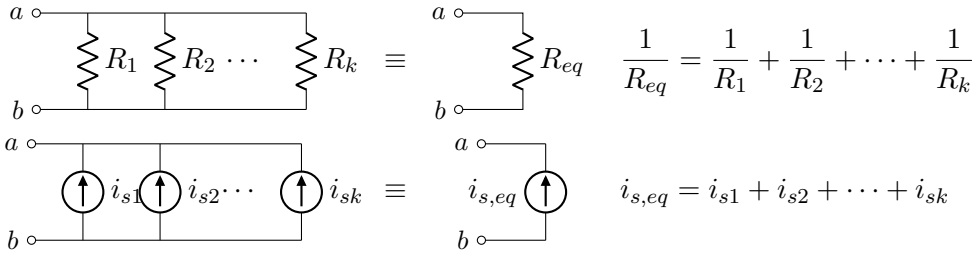
In a series connection the same current $i$ flows through every element (KCL), and the voltages add (KVL).




For the resistors: $v = R_1 i + \cdots + R_k i = R_{eq} i$. The share of $R_j$ is $v_j = (R_j/R_{eq})\, v$ (**voltage divider**). For the sources all + signs point towards $a$; a source turned the other way enters the sum with $-$.

### Parallel connection

In a parallel connection every element has the same voltage $v$ (KVL), and the currents add (KCL).



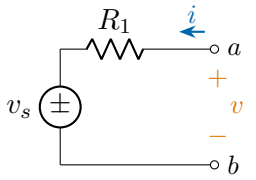

With conductances $G_j = 1/R_j$: $i = G_1 v + \cdots + G_k v = G_{eq} v$, so $G_{eq} = G_1 + \cdots + G_k$. The share of $R_j$ is $i_j = (G_j/G_{eq})\, i$ (**current divider**). Two resistors: $R_{eq} = R_1 R_2/(R_1 + R_2)$.

**Two connections that are not allowed**

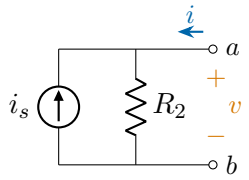
Two voltage sources with $v_{s1} \neq v_{s2}$ in parallel break KVL. Two current sources with $i_{s1} \neq i_{s2}$ in series break KCL. With equal values the connection is allowed, but the current shared by the two voltage sources (or the voltage shared by the two current sources) is not fixed.

## 5. One source and one resistor

Take a voltage source with a series resistor, and a current source with a parallel resistor. In both, $i$ enters at $a$.




$$v = R_1 i + v_s \qquad (1)$$




$$v = R_2(i + i_s) = R_2 i + R_2 i_s \qquad (2)$$

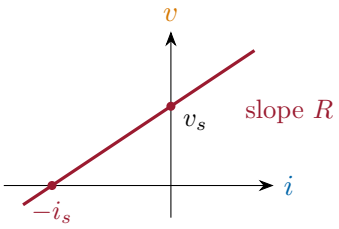
In (2), $i$ and $i_s$ both flow into the top node, so $R_2$ carries $i + i_s$. Both characteristics are straight lines in the $i$–$v$ plane. They are the same line when the slopes and the intercepts agree.

**Source transformation**

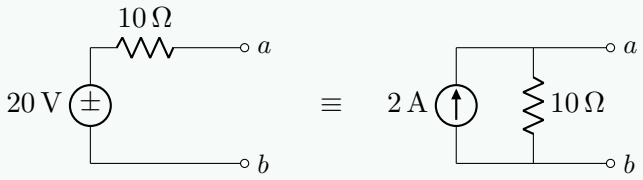
The two one-ports are equivalent if

$$R_1 = R_2 = R \qquad \text{and} \qquad v_s = R\, i_s.$$

Open circuit ($i = 0$): $v = v_s$. Short circuit ($v = 0$): $i = -i_s$.




**Example 10.3 — a source transformation**




$R = 10\,\Omega$ and $i_s = v_s/R = 20/10 = 2\,\text{A}$. Both have open-circuit voltage 20 V and short-circuit current 2 A (out of $a$).

**Example 10.4 — Example 10.2 again, by transformations**

Replace the 12 V source and the 2 Ω by a 6 A source in parallel with 2 Ω. Now 2 Ω, 6 Ω and 3 Ω are in parallel: $G_{eq} = \frac{1}{2} + \frac{1}{6} + \frac{1}{3} = 1\,\text{S}$, so $R_{eq} = 1\,\Omega$, and

$$v = 6\,\text{A} \times 1\,\Omega = 6\,\text{V}, \qquad i = 6/3 = 2\,\text{A},$$

as before. The part outside the transformed one-port sees the same $v$ and $i$. Inside it the values differ: the 12 V source carries 3 A, while the 6 A source carries 6 A.

## 6. Solving LTI resistive circuits

Series and parallel connections and source transformations simplify many circuits, but not all. Two methods write the equations of the circuit as it is, without transforming it, and work for every LTI resistive circuit:

1. **node voltage analysis** — the $n - 1$ node voltages are the unknowns;
2. **mesh current analysis** — the $b - n + 1$ mesh currents are the unknowns.

Both start from the matrices met earlier ($A$ and $M$); now the element equations are linear and can be written in matrix form too.

## Summary

- A **resistive** element has an algebraic law with no memory (resistor, dependent source, ideal transformer, ...); a **dynamic** one has a derivative ($L$, $C$). A circuit is linear, time-invariant and resistive if all its elements are. Independent sources are external inputs; they are not part of the classification. All equations of an LTI resistive circuit are linear and algebraic.
- **Substitution:** a part $N_2$ may be replaced by a voltage source equal to its port voltage, or a current source equal to its port current; nothing changes in the rest (if the solution stays unique).
- Two one-ports are **equivalent** if they have the same $i$–$v$ characteristic; the rest of the circuit cannot tell them apart.
- Series: $R_{eq} = \sum R_j$, $v_{s,eq} = \sum v_{sj}$. Parallel: $G_{eq} = \sum G_j$, $i_{s,eq} = \sum i_{sj}$.
- A voltage source $v_s$ with series $R$ is equivalent to a current source $i_s = v_s/R$ with parallel $R$.